

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4.4: Introduction to Recommender Systems

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Recommender Systems

- Recommender systems occur everywhere on the internet.
- They represent one of the most widespread applications of machine learning.

Inspiriert von deinem Browserverlauf



←

Ravensburger Puzzle-Conservier - Transparenter Puzzlekleber um Puzzles zu fixieren und...

★★★★☆ 37.635

2000+ Mal im letzten Monat angesehen

-32 % 9⁵⁶ €

UVP: 14,11 €

Erhalte es bis **Freitag, 11. Oktober**

GRATIS-Versand für Bestellungen ab 39,00 € und Versand durch Amazon



Teakschale Dekoschale Holzschale Teak Schale Schmuckschale ca. Ø 30 cm massiv Wurzelholz...

★★★★☆ 597

1000+ Mal im letzten Monat angesehen

14⁹⁵ €

Lieferung **15 Okt. - 18**

8,90 € Versand



Nariolar Puzzle Kleber Transparent mit Applikator Geeignet für...

★★★★☆ 1.438

3000+ Mal im letzten Monat angesehen

Bestseller Nr. 1 in Puzzlezubehör

9⁰⁶ €

Erhalte es bis **Freitag, 11. Oktober**

GRATIS-Versand für Bestellungen ab 39,00 € und Versand durch Amazon



envami® Weltkarte zum Rubbeln XXL mit Zubehör I Deutsch I 80x60cm I...

★★★★☆ 673

1000+ Mal im letzten Monat angesehen

Bestseller Nr. 1 in Lehrmaterial Landkarten

-15 % 21²⁴ €

Prime Angebot

24,99 €



Mangata Aufbewahrungsbox, (33x38x33cm) Aufbewahrungskorb...

★★★★☆ 724

1000+ Mal im letzten Monat angesehen

39³¹ €

Erhalte es bis **Freitag, 11. Oktober**

KOSTENFREIER Versand durch Amazon



Pufas Fensterkitt 0,500 KG

★★★★☆ 145

7 Angebote ab 5⁹⁷ €

Recommendations on the shopping website Amazon based on my browsing history.

These and the following slides are inspired by Dietmar Jannach's lecture "[Recommender Systems: Value, Methods, Measurements](#)".

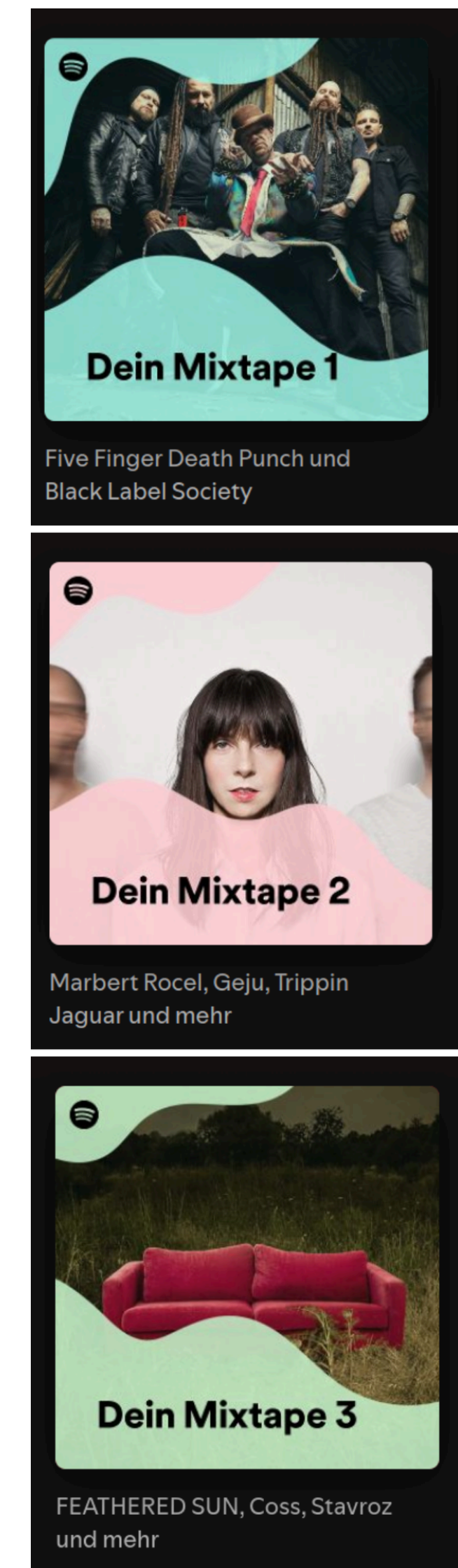
Applications of Recommender Systems

- Books
- News
- Videos
- Music
- Games
- Shopping
- Friends
- Communities
- Jobs
- Apps
- Restaurants
- Hotels
- Romantic relationships
- Stocks
- Insurance
- ...
- Wine
- ...

Why do we need Recommender Systems?

Information Overload

- We live in an **information society**. In many areas of life, more information is available than we can absorb.
- Example Spotify
 - At the beginning of 2024, there were 100 million songs on the platform.
 - Assuming each song is 3 min long, it would then take about 5 million hours to listen to all songs.
 - A typical human life has about 30,000 hours. Even if we listen to music all the time, we would only hear 0.6% of the available music.
- From the mass of available options, a small part must be **selected** → recommender systems.



Recommendations on Spotify.

Different Interests

Different stakeholders have different interests in recommender systems.

Consumers

- Find items that satisfy (long-term) needs.
- Make good decisions.
- Discover new items.

Producers / Artists

- Be found.
- Influence purchasing decisions
- Receive information about customers.
- Earn money.

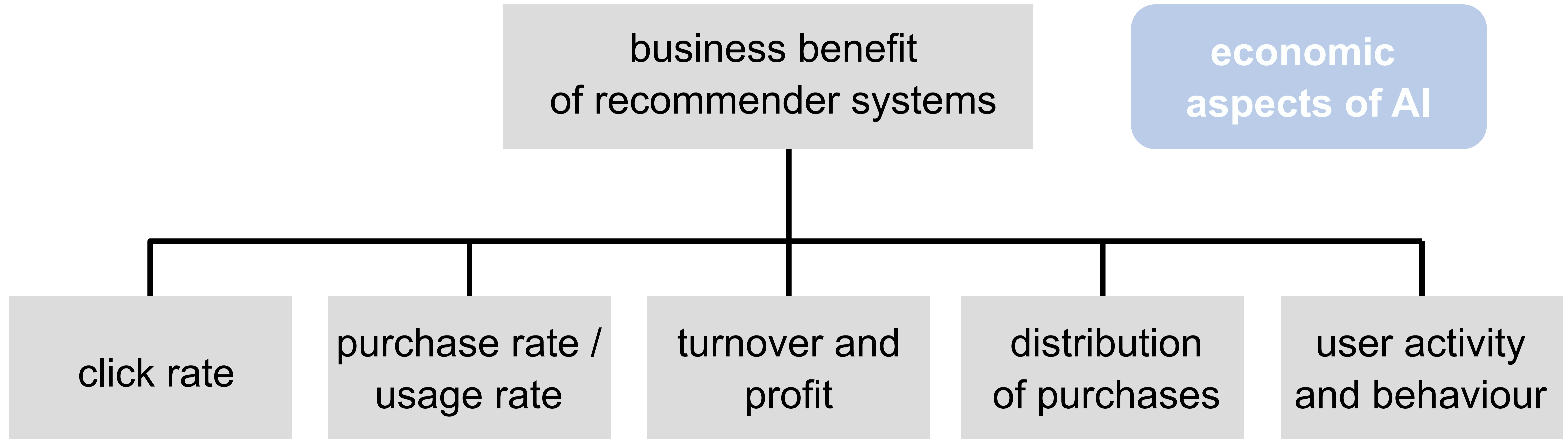
Platforms

- Increase user activity.
- Offer a value-added service
- Collect data about users.
- Show ads.
- Earn money.

How can we measure relevance concretely?

What do we measure?

- Basic assumption: the interests of consumers, producers, and platforms are compatible.
- The market regulates the efficient distribution of attention to items.



Adapted from Jannach & Jugovac [Measuring the Business Value of Recommender Systems](#), *ACM TIMS* (2019).

Click-Through Rate, Purchase Rate, Usage Rate

- **Click-Through Rate (CTR)**
 - Measures how many clicks are generated from recommendations: It is calculated as the number of clicks / number of impressions.
 - Example: News recommendations: how many articles does the user click on?
- **Purchase Rate / Usage Rate**
 - The CTR does not measure whether the user actually read/bought/used the recommended item.
 - Example: how many users actually watch a recommended video on Netflix after clicking on it?

User Behavior and Activity

- Assumption: Higher user activity leads to users **staying with the service longer**.
- This translates to users not cancelling their subscriptions, and/or users seeing more ads.
- Metrics: interactions (likes, shares, comments) with items, time spent with an item.
- For intensity and duration of activity, users and providers/platforms can have **different interests**.
Example: Dating platform.



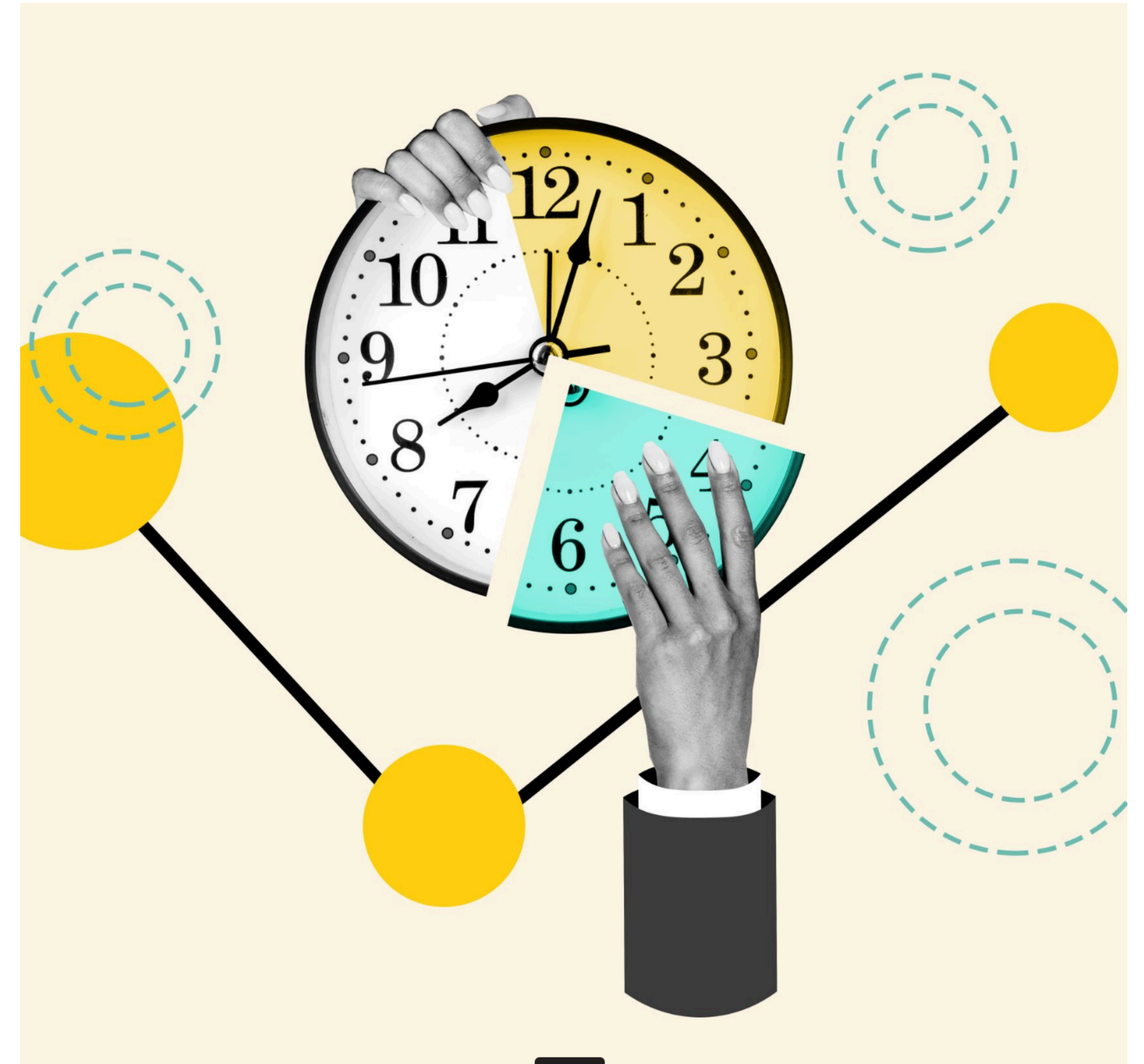
Source: [Printpeppermint.com](https://www.printpeppermint.com).

Attention Economy

- In the attention economy, users' attention becomes economic capital.
- Recommender systems that satisfy users' **short-term needs** such as entertainment can harm their **long-term interests**.
- Examples: addictive behavior, psychological health, polarization, misinformation.

ethical aspects
of AI

economic
aspects of AI



Source: [Skyword.com](https://www.skyword.com).

Summary

- Recommender systems are found practically everywhere on the internet where we are confronted with **information overload**.
- **Relevance** of individual **items** that are subject to recommendations can be measured in different ways, for example, by **click-through rate** or **purchase rate**, but also by **user activity**.
- Consumers, providers, and platforms can have **different interests**. Problems can arise if for example a recommender systems do not meet the long-term interest of consumers.